



Senior backend engineer with 10+ years building high-concurrency distributed systems. I focus on solving problems — performance bottlenecks, deployment automation, system security — not memorizing frameworks. Domain experience spans real-time gaming platforms (multi-server architectures serving multiple concurrent titles), payment processing, and enterprise APIs. My workflow is AI-augmented: day-to-day I collaborate with AI for debugging, investigation, and architecture discussions; for new feature development, I use a self-built multi-role AI loop framework to automate the Design → Code → Review → Test pipeline. I catch what AI misses — production observability, security hardening, system integrity.

Core Competencies

System Modernization & Multi-Tenancy

Solo-led full platform rewrite from Lua/PHP/Java to Go with config-driven multi-tenant architecture, replacing 13 per-platform release branch forks with a single codebase. Designed Proto2/Proto3 dual-compatible serialization layer for seamless legacy client migration.

Golang, gRPC, Protobuf, Redis, MySQL, MongoDB

High-Concurrency Backend Architecture

Designed multi-server distributed systems (Gate/Lobby/Game) across 10+ repositories and multiple platforms. Built reusable frameworks that cut new title development cycles by over 50%.

Lua/Skynet, Golang, TCP/WebSocket, gRPC

Database & Cache Optimization

Diagnosed full-collection scans on ~2.7M records/day and designed compound indexes. Restructured Redis key architecture to eliminate excessive SCAN operations.

MongoDB, Redis, MySQL, PostgreSQL

DevOps & CI/CD

Automated Docker image builds to Harbor registry, designed Helm chart deployments on Kubernetes, set up GitLab Runners and remote SSH deployments.

Kubernetes, Docker, Helm, GitLab CI/CD

Security & Data Integrity

Investigated Redis data tampering incidents, analyzed full attack chains. Designed HMAC-SHA256 data signing mechanism to architecturally prevent direct Redis manipulation that bypasses application-layer validation.

Employment History

- **Senior Backend Engineer** · Tyche Tech Co, Ltd Jan. 2021 — Present
3 game server projects (Lua/Skynet, Golang). Fixed reward-claim race condition, MongoDB optimization on 2.7M records/day. Redis architecture redesign. Security incident investigation & AES encryption. CI/CD pipeline with Docker/Harbor/Helm.
- **Senior Backend Engineer** · Terpro Mobile Co, Ltd Jun. 2019 — Dec. 2020
Built reusable slot game framework (FreeGame, Lightning, PickGame). Kubernetes/Helm deployment.
- **Senior Backend Engineer** · WistronITS Corp. Stationed: Taiwan Mobile Sep. 2016 — May 2019
Led 2 engineers. Built 5 micro-services (OpenId, Product, Payment, Stock, Kafka). Jenkins/Nexus CI toolchain.
- **Software Engineer** · Kangda Info Corp. Jul. 2015 — Aug. 2016
- **Software Engineer** · BenQ Corp. Nov. 2013 — Feb. 2015
- **Software Engineer** · Fashion Intelligence Co, Ltd (Open-Life) Nov. 2011 — Oct. 2013

Freelance & Side Projects

- **PGS Game Server (solo)** Dec. 2022 — Apr. 2023
- **Yapro car tax refund management (2 members)** Jan. 2021 — Mar. 2021
- **Muxin CarePro CMS (5 members)** Nov. 2013 — Mar. 2015

Education

Bachelor of Electrical Engineering — National Kaohsiung University of Science and Technology(NKUST), Taiwan Sep. 2006 — Jun 2010

AI-Augmented Development

Day-to-day I collaborate with AI through prompting — debugging, investigation, architecture discussions. For new feature development, I switch to a self-built loop framework that lets AI iterate autonomously while I handle architecture design and final acceptance.

- Built 4x (github.com/ggwhite/4x) — an open-source multi-role AI dev framework: Designer → Coder → Reviewer → Tester → Deep Reviewer → Acceptor, with cross-role oversight, automatic FAIL-retry cycles, batch execution, and structured reporting
- Architect-first workflow: I make architecture decisions (module boundaries, protocols, security model), then use AI agents for implementation — catching production blind spots that AI overlooks, like logging coverage and security hardening
- Engineering guardrails for AI: harness configuration (CLAUDE.md / AGENTS.md / coding-rules) ensures AI follows project conventions; external memory (learnings.json) prevents repeated mistakes
- Cross-stack delivery: 10+ years across Golang, Java, Lua, JavaScript — combined with AI collaboration, enables rapid context-switching across any tech stack